INDIVIDUAL REFLECTION

Mini Use Case 02 – Credit Pre-Screening

Calibo AI Academy | Part 2 – KLU Batch | Phase 2 | Supervised Learning Part 1

Team Member	Sai Surchit Das
Role	Data Analyst – Credit Risk Analysis (Team Lead)
Contribution	Step 5.2: Understand Approval Drivers and Step 5.3: Engineer Affordability Features

1. Project Overview

Our team worked on Mini Use Case 02 in the BFSI – Retail Lending (Personal Loans) domain. The objective was to understand how the credit team made historical loan decisions and prepare meaningful affordability features for supervised Machine Learning. The dataset contains 12,000 loan applications with information about applicant demographics, employment, income, existing EMI, credit history, loan details and historical decisions.

2. My Role and Contribution

Step 5.2 — What drives approval today? I analysed historical approval rates across credit-score bands and employment types. The analysis showed a sharp difference by credit score: applicants with scores from 600–649 had an approval rate of about 4.3%, while the 650–699 band had about 32.8%, and applicants scoring 750 or above had an approval rate of roughly 82%. This suggests that 650 is an important point in the bank's historical decision pattern. Applicants with no credit score (new-to-credit) had an approval rate of about 53.9%, showing that missing credit history should not automatically be treated as a decline. Approval also varied by employment type: about 58.4% for salaried applicants compared with 34.9% for gig/contract applicants.

Step 5.3 — Engineer affordability features. I created three features to describe applicants' repayment burden and credit-history status. FOIR was calculated as (existing EMI + proposed EMI) divided by monthly income. Loan-to-income was calculated as loan amount divided by annual income (monthly income × 12). `is_ntc` was set to 1 when the applicant had no credit score and 0 otherwise. These features help represent affordability and distinguish applicants who are new to credit from those with an established credit record.

3. Key Findings

Measure	Historical result	Interpretation
Credit score 600–649	4.3% approved	Approval was very uncommon in this band.
Credit score 650–699	32.8% approved	Approval rises substantially after 650.
Credit score 750+	About 82% approved	Higher scores were associated with much higher approval.
No credit score / NTC	53.9% approved	These applicants need separate consideration, not automatic rejection.

Measure	Historical result	Interpretation
FOIR below 30%	65.4% approved	Lower repayment burden was associated with higher approval.
FOIR 50–55%	31.7% approved	Approval had fallen as repayment burden increased.
FOIR above 55%	15.5% approved	Very high FOIR was associated with low approval.

4. Interpretation and Recommendation

The analysis suggests that both credit history and repayment affordability are important in historical lending decisions. The approval rate drops from 65.4% for applicants with FOIR below 30% to 15.5% for applicants above 55%. Based on this pattern, a proposed policy review point could be around 50–55% FOIR, while recognising that FOIR should be considered alongside credit score, employment, and other relevant information rather than used as the only decision rule. New-to-credit applicants should be assessed carefully because their approval rate is about 53.9%, despite having no credit score.

5. Final Summary

My contribution covered Steps 5.2 and 5.3. I analysed approval patterns by credit-score band and employment type, identifying a strong approval increase at higher credit scores and lower approval among gig/contract applicants compared with salaried applicants. I then engineered FOIR, loan-to-income and the new-to-credit indicator to capture repayment burden and credit-history status. The results showed that approval rates fall sharply as FOIR rises, particularly above 50–55%. These findings help explain historical credit decisions and provide useful features for the next stage of the pre-screening model. They describe past decisions and should not be treated as proof that any single feature causes an outcome.